

Summary | Third-year Ph.D. student building e-graph-based tools for hardware design automation, spanning arithmetic datapath synthesis, high-level synthesis, and formal verification. Research sits at the intersection of programming languages, machine learning, and hardware design, targeting scalability, correctness, and optimization in next-generation design flows. Ph.D. research funded directly by AMD as a Graduate Applied Researcher. Australian citizen on an F-1 visa; CPT-eligible for U.S. internships.

Education

Georgia Institute of Technology

Aug 2024 – May 2028 (Expected)

Ph.D. in Electrical and Computer Engineering — 4.0 GPA

Advised by Prof. Callie Hao, SHARC Lab

- Coursework: SAT/SMT Solvers, Programming Languages, HW–SW Co-Design for ML, FPGA Programming

Imperial College London

Oct 2020 – Jun 2024

M.Eng. in Electronic and Information Engineering — First Class Honours (4.0 GPA equivalent)

Industry Experience

Numerical Hardware Engineer Intern, AMD

May 2026 – Aug 2026

- Built an e-graph rewriting tool over fully mapped standard-cell netlists, reducing cell count and area; measured figures under NDA
- Scaled the flow to production netlists of over 4 million cells, now under AMD IP synthesis evaluation
- Extended the ILP extraction to stay tractable at netlist scale, with delay constrained, free, or optimized

Numerical Hardware Engineer Intern, AMD

May 2025 – Aug 2025

- Designed integer and constant multiplier architectures that improve both minimum delay and area
- Designed an e-graph multiplier framework spanning array-generation and reduction, enabling hybrid architectures
- Developed methods to generate multipliers optimized for input properties such as arrival time and toggle rates

Numerical Hardware Engineer Intern, Intel

Jun 2024 – Aug 2024

- Automated mixed-precision floating-point multiplier optimization in Rust with e-graphs, improving PPA up to 10%
- Investigated glitch power minimization in hardware designs using integer linear programming

FPGA Engineering Co-op, Quantum Motion

Apr 2023 – Sep 2023

- Designed a bespoke FPGA signal generator for high-speed qubit feedback, with a PYNQ/QICK Python interface
- Programmed an RP2040 microcontroller in C to communicate over TCP/IP on Ethernet connections

Analog and Digital IC Validation Intern, Quantum Motion

Jul 2022 – Sep 2022

- Validated ring oscillators on silicon, analyzing transistor properties at room and cryogenic (2K) temperatures

Research Experience

EGGROLL: Reinforcement Learning for E-Graph Exploration

Jan 2026 – Present

Georgia Institute of Technology and AMD — first author, in preparation; invited demo, ARITH 2026

- Building a generalized RL framework to address the scalability limits of equality saturation
- Formulating equality saturation as a decision problem: e-graph as state, rewrite as action, sparse cost reward
- Matching or improving on expressions from exhaustive equality saturation with far smaller e-graphs
- Designed to apply to any e-graph framework given a language, rewrite set, analyses, cost function, and extraction method; currently driving a netlist rewriting flow, guiding exploration away from e-graph blow-up on large designs

Lemonade: Automated HLS Design Modularization using E-Graphs

Mar 2025 – Present

Georgia Institute of Technology — first author, under review at ASPLOS 2027

- Built an MLIR-based framework that automatically discovers reusable hardware modules across structurally distinct HLS programs, combining e-graphs, anti-unification, and polyhedral analysis
- Introduced polyhedral loop-mapping analysis deriving the minimal set of sound loop transformations that expose shared modules, pruning an 8,716M-candidate space to 4,096 designs (0.00005%)
- Designed decoupled hardware-oriented cost functions — a hardware impact cost for module extraction and a performance cost for implementation — reusing 38.1% of synthesized hardware
- Composed with ScaleHLS for geo-mean 47.8% lower area at iso-delay and 53.8% fewer cycles at iso-area

ForgeBench: An HLS Design Generator of ML Test Suites for Next-Generation HLS Tools

Apr 2025 – Present

Georgia Institute of Technology — co-first author, under review at ACM TRETS

- Built a generator that composes synthesizable HLS C/C++ from an extensible ML kernel library, so HLS tools can be developed against current ML workloads rather than legacy fixed benchmarks
- Generated full model-level designs for the ResNet family and LLaMA-3.1-8B, including a complete 32-layer LLM project spanning prefill and decode, KV-cache, RoPE, and attention, validated against a NumPy golden reference
- Demonstrated scale with a 12,912-design suite across GEMM, DNN, and LLM, 2,711 place-and-routed on a ZCU102

SpecScore: Quantifying Readability of HLS-Generated RTL Using LLMs

May 2026 – Present

Georgia Institute of Technology — co-first author, in preparation

- Proposed the first automated, annotation-free RTL readability metric: the cosine similarity between an HLS C++ specification and its tool-generated RTL in a joint LLM embedding space
- Introduced four strategies for embedding RTL exceeding model context windows, including hierarchical module summarization, validated at 0.95 mean reciprocal rank on PolyBench, MachSuite, and CHStone

Formal Verification of Hardware Arithmetic in Lean

Aug 2025 – Dec 2025

Georgia Institute of Technology

- Built a Rust-to-Lean 4 flow verifying RTL arithmetic datapaths against a bitvector DSL spec, translating Verilog via Yosys and AIGER and discharging equivalence with `bv_decide`/CaDiCaL
- Designed a type-explicit arithmetic DSL forcing a bitwidth on every wire, $3.64\times$ more concise than equivalent RTL
- Implemented canonical bitwidth reconciliation mapping hardware's $n, m \rightarrow k$ operators onto Lean's BitVec

OptiMult: Multiplier Optimization via E-Graph Rewriting

Nov 2022 – Jun 2023

Imperial College London, with Intel and UCLA — first author, ASILOMAR 2023

- Built an e-graph rewriting tool in Rust on egg that explores equivalent multiplier architectures as local equivalence-preserving rewrites and emits synthesizable Verilog
- Introduced a dual row and column representation of AND arrays, and a two-phase optimization with separate rewrite sets and delay cost models per phase to keep gate-level search tractable
- Cut latency by up to 46% on squarers and 9% on multipliers against a commercial synthesis tool at TSMC 5nm, with every design formally verified by equivalence checking

OptiINN: Breaking the Interpretability–Efficiency Trade-off for DNNs on GPUs

Oct 2023 – Jun 2024

Imperial College London — M.Eng. thesis

- Designed a prototype-based interpretable architecture that recasts prototype distance as GPU-optimized convolutions with hard-sigmoid activations, for full cuDNN and TensorRT support
- Built a toolchain compiling pretrained CNNs into optimized interpretable TensorRT engines, with principled quantize/dequantize node placement and quantization-aware training for INT8
- Showed prototype interpretability resists quantization, I-Score moving under 0.03 from FP32 to INT8 while latency fell up to 84.4%: MobileNet-V3 runs interpretably in 2.10 ms against 8.35 ms

Technical Skills

Languages: Rust, Python, C/C++ , SystemVerilog/Verilog, Lean

EDA tools: Design Compiler, Fusion Compiler, JasperGold, VCS, Vitis HLS, Catapult HLS, Yosys

Hardware flows: Logic synthesis, formal equivalence checking, static timing analysis, place and route, RTL design, standard-cell and mapped-netlist flows, FPGA prototyping (PYNQ, QICK)

Methods: E-graphs and equality saturation (egg), MLIR, ILP (Gurobi), SAT/SMT, reinforcement learning, PyTorch

Publications

Lemonade: Automated HLS Design Modularization using E-Graphs

First author.

Under Review, ASPLOS 2027

ForgeBench: An HLS Design Generator of ML Test Suites

Co-first author. Extends our FCCM 2026 poster; FPT 2026 Journal Track.

Under Review, ACM TRETs 2026

[Preprint](#) [Code](#)

Hardware Accelerator Generation for Cryptographic Primitives

Third author.

ASAP 2025

[IEEE Xplore](#)

OptiMult: Multiplier Optimization via E-Graph Rewriting

First author.

ASILOMAR 2023

[IEEE Xplore](#)

Honors & Awards

AMD Graduate Applied Researcher

Ph.D. research funded by an AMD gift and a sponsored research agreement

Oct 2025 – May 2028

ACM/IEEE DAC Young Student Fellow

2025

Dean's List, Imperial College London

Top 10% of cohort in third year, top 5% in fourth

2023 and 2024

Teaching, Mentoring & Service

Graduate Teaching Assistant, CS 4240 Compilers

Georgia Tech, MIPS Lecture

Spring 2025, Fall 2025

Teaching Assistant, ELEC40013 Mathematics

Imperial College London

Fall 2022

Reviewer, IEEE Transactions on Computers

2026

Mentor to an M.Sc. student, work published at ASAP 2025

2025